YASHAS B D

Embedded Firmware Engineer

yashasbdvviet@gmail.com | linkedin.com/in/yashasbd

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 3 years of experience in low-level firmware development for multi-processor embedded platforms. Specialized in bootloader development, device driver implementation, DMA optimization, processor-to-processor communication, and hardware-software integration. Demonstrated success in improving system performance through high-throughput QSPI driver optimization, developing reliable firmware update mechanisms, and resolving complex firmware issues across mission-critical embedded systems. Proficient in Embedded C, ARM Cortex-M architectures, peripheral interfaces, and advanced debugging techniques, with a strong foundation in building robust and maintainable embedded software solutions.

TECHNICAL SKILLS

- **Programming Languages:** C, Python.
- **Embedded Systems:** Bootloader Development, BSP Development, Device Drivers, Bare Metal Programming, Firmware Architecture.
- **Peripherals & Protocols:** QSPI, SPI, I2C, UART, DMA.
- **Debugging:** JTAG, SWD, Logic analyzers, Oscilloscope, Memory Analysis.
- **Software Tools:** Git, SVN.

PROFESSIONAL EXPERIENCE

Accord Software & Systems Private Limited, Bengaluru, India

Systems Engineer *August 2023 – Present*

Project: Triple antenna

- Engineered multi-stage bootloaders for 4-processor platforms, transitioning from UART to SPI upgrades, and reducing deployment time by 40
- Optimized high-speed DMA-based QSPI drivers, achieving 95% CPU overhead reduction and 3x throughput improvement for mission-critical data operations.
- Designed and implemented SPI Master drivers with GPIO arbitration for shared-bus architectures, enabling reliable communication across multiple peripherals without contention.

Project: MHDGR

- Engineered interrupt-driven I2C communication routines for NVMM data persistence, ensuring reliable storage operations with less than 1ms latency and zero data loss,
- Performed advanced debugging and defect resolution using JTAG/SWD interfaces with real-time code tracing, reducing firmware debug cycles by 50%,
- Validated peripheral interfaces and improved signal integrity across high-frequency communication buses through cross-functional collaboration with hardware design teams.

Project: UHF

- Programmed and configured RF IC modules for wireless communication links, implementing error correction and retry mechanisms for 99.9% transmission reliability.

Project: Handheld

- Developed peripheral drivers and contributed to the integration of system-level firmware in multiple projects, ensuring reliable hardware-software interface.
- Built test frameworks to validate communication protocols (SPI, I2C, UART, CAN) and peripheral functionality across multiple hardware platforms, improving test coverage and reducing validation time.
- Diagnosed and resolved complex multi-layer firmware issues using oscilloscopes, logic analyzers, and software debugging tools, cutting down hardware bring-up and integration time.

L&T Technology Services
Engineering Intern *September 2022 – October 2022*

- Designed and deployed a Python GUI application for real-time PFC parameter monitoring, integrating serial communication for live hardware data capture and implementing dual-mode visualization — numerical readouts alongside dynamic time-series plots — for in-depth system analysis.

EDUCATION

Vidya Vikas Institute of Engineering and Technology
B.E (ECE) CGPA: 8.69/10 *August 2019 – July 2023*
Relevant Coursework: Embedded Systems, Microprocessors & Microcontrollers, Digital Signal Processing, VLSI Design

ACADEMIC PROJECTS

Instinctive Invoicing Bot

- Designed a trolley-mounted automated self-checkout system enabling customers to scan, bill, and pay for items via RFID before exiting the store.
- Integrated barcode scanning with real-time LCD display of product details and pricing, paired with RFID-based payment processing for seamless checkout.
- Architected a multi-threaded firmware solution using a state machine approach to manage concurrent Wi-Fi connectivity and real-time sensor polling.

IoT Based Pet Feeder

- Built a Wi-Fi enabled remote pet feeding system using NodeMCU, allowing users to control feeding schedules, monitor food levels, and manage pet health from anywhere via the Blynk platform.
- Developed and validated NodeMCU firmware integrating Wi-Fi connectivity, cloud-based monitoring, and actuator control, ensuring reliable remote operation through the Blynk cloud interface.

CERTIFICATIONS

Programming With C And C++ **December 2020 – January 2021**
Internshala Trainings

- Successfully completed an eight weeks online certified training on Programming with C and C++.
- The training consisted of Getting Started With Programming in C, Diving Into C Programming, Fundamentals of Object Oriented Programming Using CPP, Diving into CPP Programming